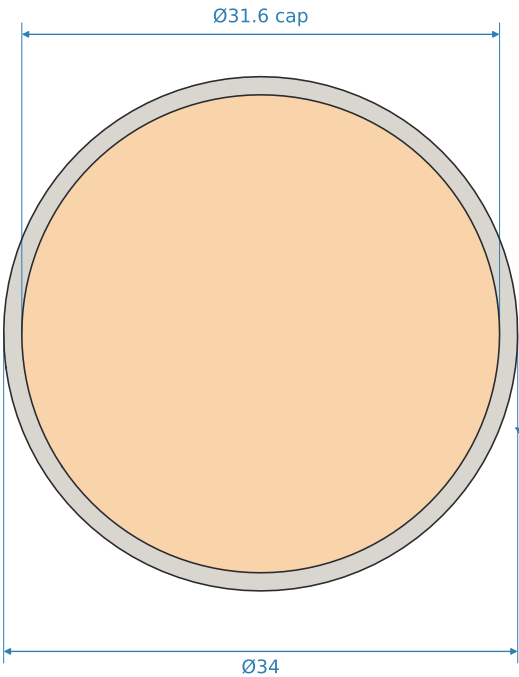
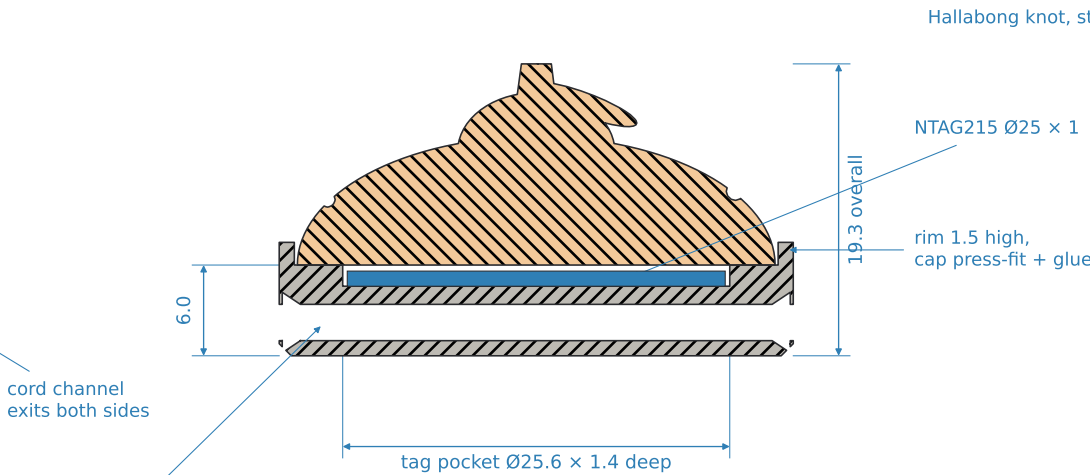


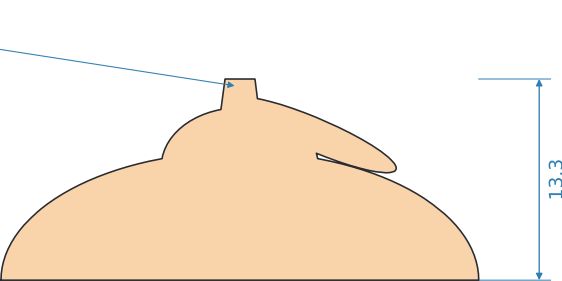
Charm, top view
scale 2:1



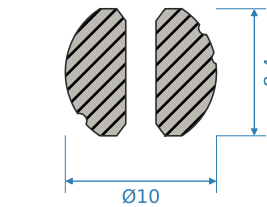
Charm assembly, section A-A
scale 2:1



Cap, front view
scale 2:1

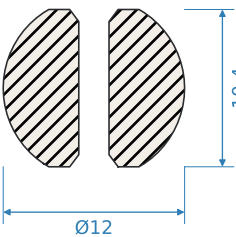


Basalt bead, section



hole Ø2.0 · dark grey / black PLA

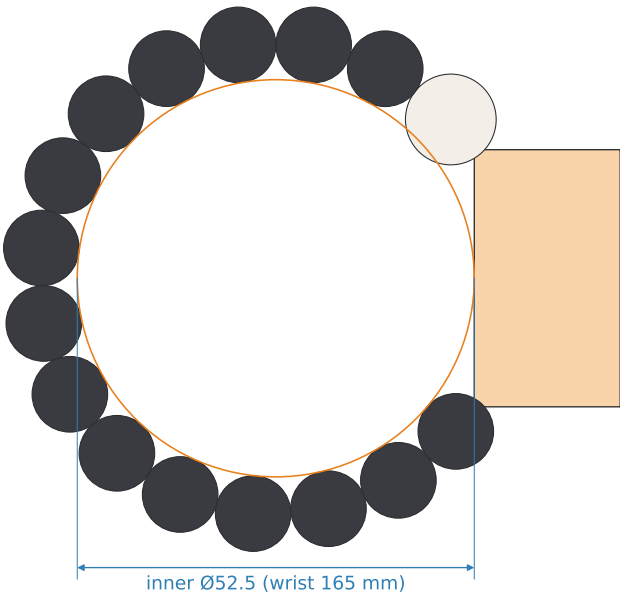
UV bead, section



hole Ø2.0 · UV colour-change filament

Bracelet layout, size M (top view)
scale 1:1

charm faces out,
flat base on the wrist



Sizes (elastic stretch fit)

S: wrist 150 mm → 13 basalt beads + 1 UV bead + charm
M: wrist 165 mm → 15 basalt beads + 1 UV bead + charm
L: wrist 180 mm → 16 basalt beads + 1 UV bead + charm
Elastic cord Ø1.0 mm clear, about 25 cm per band;
surgeon's knot hidden inside the charm channel.

Parts and printing

Charm base PLA basalt grey 0.12 mm layers, 100 % infill
Charm cap PLA tangerine flat side down; supports off
Basalt bead PLA black/stone hole vertical, 20 % infill
UV bead UV colour-change PLA (white → colour in sun)
NFC tag NTAG215 Ø25 mm sticker, placed before the cap
Glue a drop of CA glue on the rim before pressing cap

Assumptions

- Tag Ø25 × 1.0 mm (common NTAG215 coin sticker).
- Read distance through the 1.5 mm cap floor: the tag faces the reader when the charm is tapped face-down.
- Pocket and rim gaps 0.2–0.3 mm per side; tune `fit\_clearance` in models.py for your printer.
- The beads are copies of one STL: print 13 / 15 / 16.
- Real volcanic basalt beads (10 mm, Ø2 hole) can replace the printed ones with no other change.

Wristband

Hallabong NFC charm · basalt beads · UV bead

Jeju wish-band · 2026 GLOBAL UNIFORCE · KMUTNB

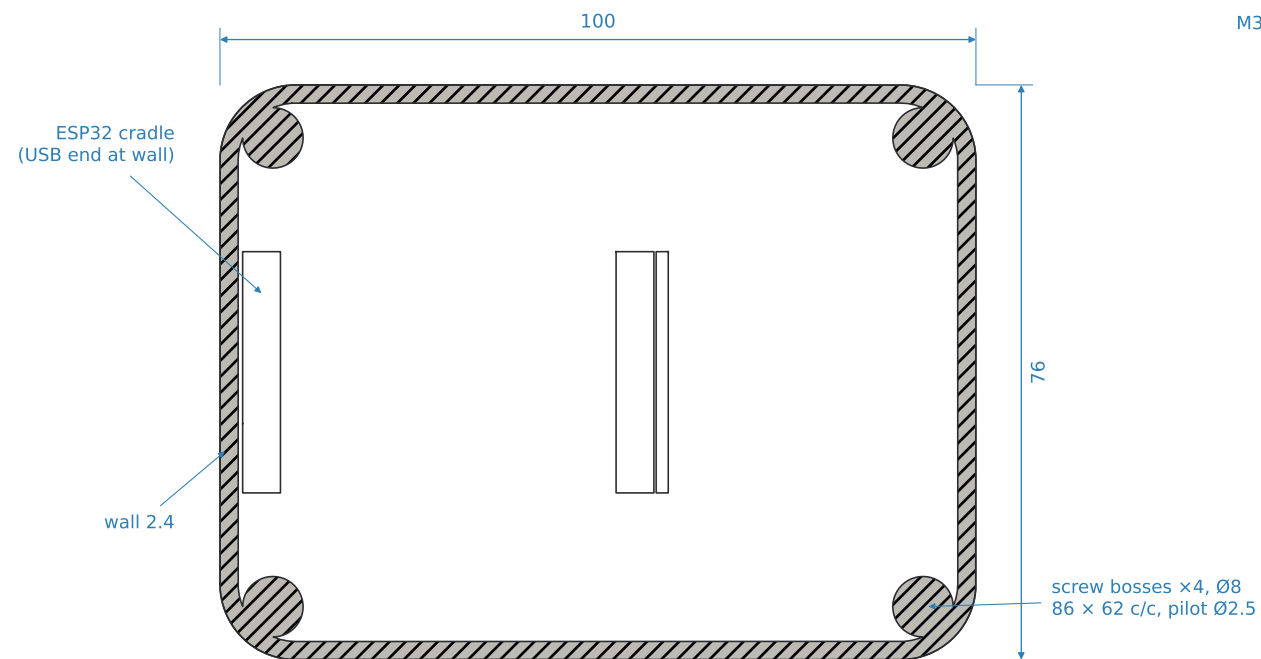
Sheet 1/2

Units: mm

A3 · print at 100 %

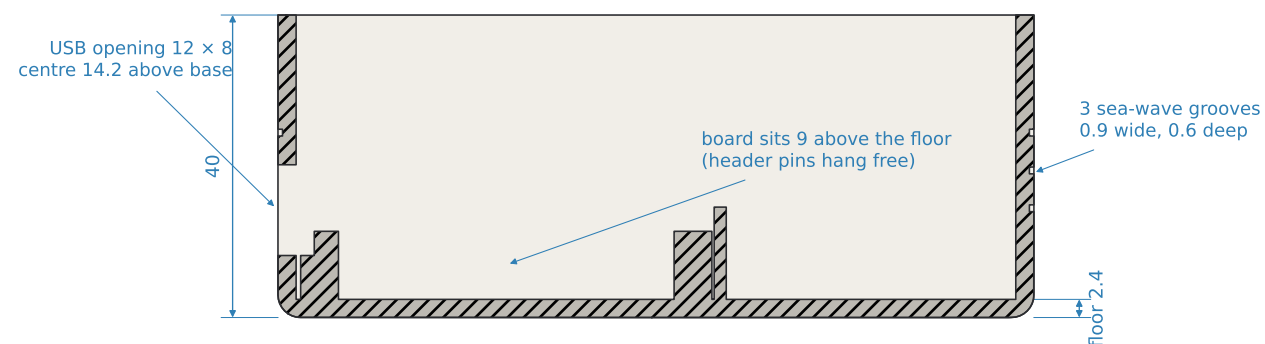
Base, section at 8 mm (top view)

scale 1:1



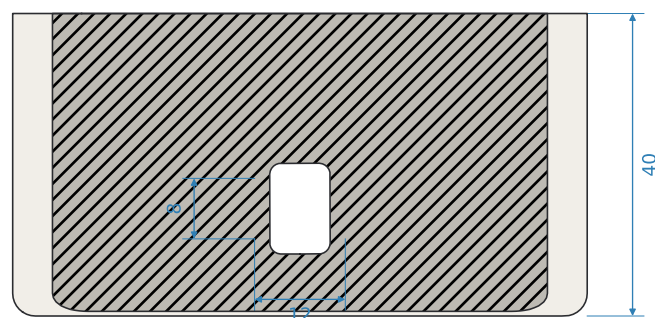
Base, section B-B (front)

scale 1:1



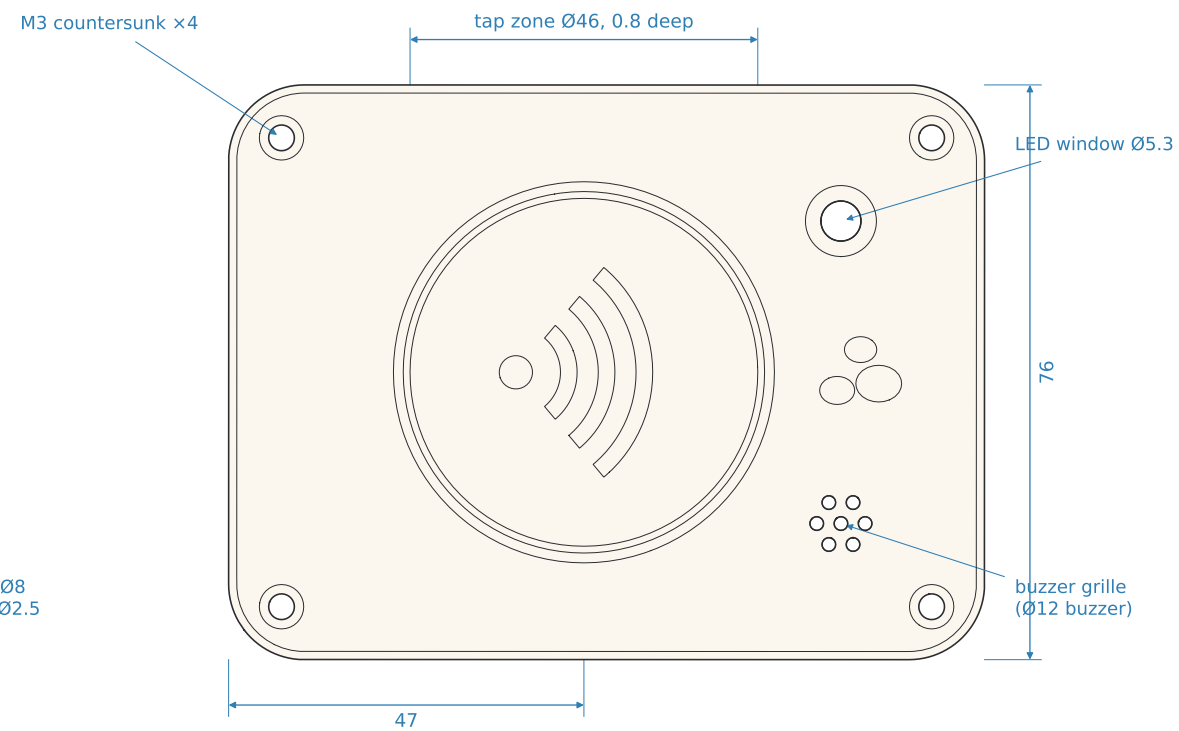
Base, left side (USB end)

scale 1:1



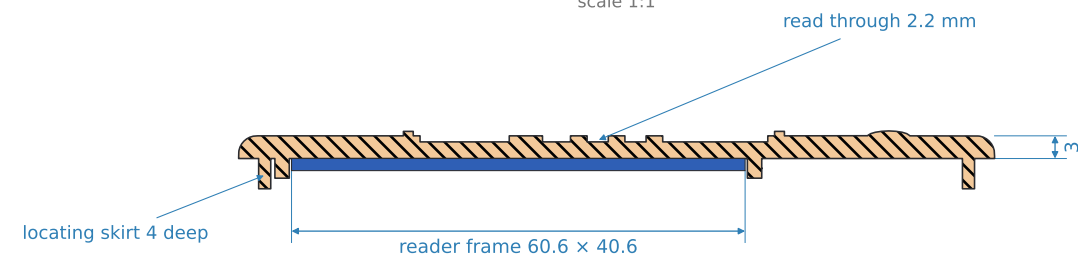
Lid, top view

scale 1:1



Lid, section C-C with RC522

scale 1:1



Components (sizes assumed, check yours)

ESP32 DevKitC V4	54.4 × 27.9 mm, pins down
RC522 board	60 × 40 mm (food / place / activity)
PN532 V3 board	42.7 × 40.4 mm (admin desk)
Buzzer	Ø12 × 9.5 mm, GPIO 4
LED	5 mm, GPIO 2
Screws	M3 × 8 self-tapping, countersunk × 4
Feet	Ø10 rubber bumpers × 4

Printing and assembly

Base PLA/PETG basalt grey, open side up, no supports (USB opening bridges 12 mm).

Lid PLA/PETG cream, flat top on the bed (STL is already flipped), supports off.

0.2 mm layers, 3 walls, 15–20 % infill.

Reader: stick the RC522 in the frame with foam tape, antenna end under the tap circle; same frame takes the PN532 for the admin desk.

Check-in station

Enclosure base + lid for ESP32 + RC522 / PN532

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Sheet 2/2

Units: mm

A3 · print at 100 %